

MATAD, Mongolia

WMO: 443140

Lat: 46.9550N

Lon: 115.2964E

Elev: 907

StdP: 90.89

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -27.4	(d) -33.2	(e) 0.2	(f) -29.3	(g) -31.0	(h) 0.2	(i) -27.0	(j) 13.6	(k) -16.9	(l) 12.2	(m) -16.3	(n) 2.8	(o) 320	(p) 0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.2	(c) 32.5	(d) 17.6	(e) 30.2	(f) 16.8	(g) 28.2	(h) 16.3	(i) 19.9	(j) 28.6	(k) 18.7	(l) 26.1	(m) 17.8	(n) 24.6	(o) 4.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 17.5	(b) 14.0	(c) 22.2	(d) 16.5	(e) 13.1	(f) 21.2	(g) 15.5	(h) 12.3	(i) 20.7	(j) 61.2	(k) 28.6	(l) 57.3	(m) 26.2	(n) 54.2	(o) 24.3	(p) 25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	10.5	9.9	-33.0	35.3	2.1	2.9	-34.5	37.4	-35.7	39.1	-36.9	40.7	-38.4	42.8
			-33.0	21.8	2.3	1.4	-34.7	22.8	-36.0	23.6	-37.4	24.4	-39.0	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-18.9	-14.9	-6.8	3.5	11.6	18.0	21.0	18.7	12.3	2.4	-9.2	-16.6
	DBStd	14.91	4.75	5.71	6.33	5.39	4.81	4.16	3.13	3.60	4.77	5.34	6.55	4.78
	HDD10.0	4029	896	697	522	206	38	1	0	1	29	241	576	823
	HDD18.3	6217	1154	930	780	446	216	56	8	38	188	493	826	1082
	CDD10.0	1050	0	0	0	10	86	239	341	269	99	6	0	0
	CDD18.3	198	0	0	0	0	7	44	91	48	8	0	0	0
	CDH23.3	2266	0	0	0	5	141	572	903	533	111	1	0	0
	CDH26.7	784	0	0	0	1	43	208	331	177	26	0	0	0
Wind	WSAvg	4.4	4.3	4.3	4.6	5.2	5.3	4.1	3.7	3.6	4.1	4.4	4.2	4.6
Precipitation	PrecAvg	299	3	3	7	13	25	56	85	58	26	11	7	6
	PrecMax	564	10	8	13	35	73	103	241	137	56	26	23	13
	PrecMin	165	1	1	1	2	5	4	22	22	6	2	1	2
	PrecStd	85	2	2	3	8	14	24	47	28	13	6	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.5	3.2	14.2	23.2	31.2	34.9	35.8	34.4	29.7	21.1	9.5	-1.7
		MCWB	-6.5	-1.1	4.7	9.7	13.5	17.8	19.8	19.8	15.6	10.2	2.8	-4.0
	2%	DB	-7.4	-0.8	9.7	19.4	27.0	31.3	32.7	30.7	26.8	17.1	6.0	-5.3
		MCWB	-9.0	-3.9	2.2	7.8	12.2	16.0	18.2	18.0	14.5	8.2	1.0	-7.0
	5%	DB	-9.7	-3.6	6.1	16.2	24.1	29.0	30.3	28.3	23.9	13.7	2.9	-7.6
		MCWB	-11.0	-5.9	0.2	6.0	11.5	15.5	18.0	16.9	13.4	5.9	-1.1	-9.0
	10%	DB	-11.8	-5.9	3.1	13.5	21.5	26.6	28.2	26.2	21.2	11.1	0.3	-9.6
		MCWB	-12.9	-7.8	-1.5	4.8	10.0	15.0	17.4	16.0	12.1	4.6	-2.7	-10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.2	-0.7	4.9	10.5	15.1	19.7	21.8	21.5	17.8	11.2	3.5	-3.4
		MCDB	-4.6	2.6	13.2	22.3	26.3	30.8	30.4	31.0	25.2	20.1	9.4	-1.0
	2%	WB	-8.8	-3.5	2.5	8.3	13.4	17.7	20.2	19.2	15.5	8.4	1.5	-6.6
		MCDB	-7.1	-0.4	9.5	18.3	24.2	27.0	28.8	26.8	23.4	16.2	6.6	-4.8
	5%	WB	-10.6	-5.5	0.6	6.5	12.0	16.8	19.2	18.1	14.2	6.6	-0.3	-8.5
		MCDB	-9.3	-3.2	6.2	15.5	22.5	25.4	27.0	24.8	21.7	13.0	3.8	-7.1
	10%	WB	-12.6	-7.3	-1.2	4.9	10.7	16.0	18.4	17.3	12.9	4.9	-1.9	-10.1
		MCDB	-11.6	-5.3	3.2	13.2	20.5	24.1	25.2	23.7	20.1	10.6	1.2	-8.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.5	11.8	13.0	13.7	12.2	11.2	12.0	12.8	11.8	10.1	8.4
		MCDBR	11.6	12.9	15.4	17.3	17.7	15.6	14.5	15.3	16.1	15.4	13.0	9.7
		MCWBR	10.0	10.2	9.9	9.7	8.5	6.4	5.5	6.5	7.9	9.0	8.8	8.1
	5% WB	MCDBR	11.2	12.6	15.0	15.9	15.6	13.0	12.0	12.2	14.4	14.6	12.5	9.6
		MCWBR	9.9	10.2	9.8	9.4	8.3	5.8	5.4	5.7	7.7	8.9	8.7	8.2
Clear-Sky Solar Irradiance	taub	0.253	0.274	0.315	0.373	0.442	0.454	0.415	0.377	0.336	0.307	0.273	0.257	
	taud	2.322	2.291	2.255	2.144	1.982	2.066	2.254	2.328	2.381	2.385	2.355	2.297	
	Ebn at Noon	849	900	906	877	824	816	845	866	875	848	813	792	
	Edh at Noon	72	93	114	141	173	160	131	117	100	85	70	66	
All-Sky Solar Radiation	RadAvg	2.06	3.23	4.64	5.74	6.46	6.45	6.17	5.75	4.66	3.27	2.11	1.61	
	RadStd	0.12	0.19	0.19	0.26	0.34	0.46	0.28	0.31	0.25	0.19	0.11	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	-2.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page